

Lecture 1:

Characteristics and Types of Computers

Learning Outcomes

- Define a computer in your own words and distinguish clearly between data and information.
- List and explain the characteristics (advantages) of computers, with a real-life example for each.
- List and explain the limitations (disadvantages) of computers.
- Describe the classification of computers by technology used, by size & capacity, and by purpose.

1 What Is a Computer? Data vs Information

A **computer** is an electronic device that receives input, stores or processes the input as per user instructions, and provides the output in a desired format.

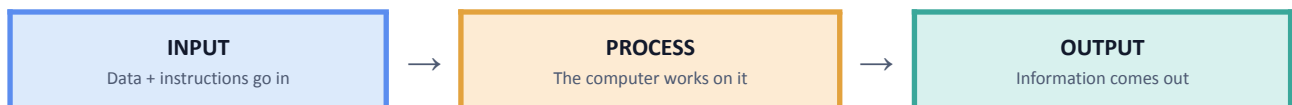


Fig 1.1 — The basic working cycle of every computer.

Data vs Information

Data is defined as the collection of raw facts and details — text, figures, observations, or symbols — that have not yet been processed. It is unprocessed, and on its own does not convey any specific meaning. The word “data” comes from the Latin *datum*, meaning “something given.”

Information is data that has been processed, organised, and structured so that it is meaningful and useful. Processing removes unnecessary detail and adds context, which reduces uncertainty and improves understanding. The word “information” comes from the Latin *informare*, meaning “to give form to.”

Point	Data	Information
What it is	Raw, unprocessed facts and figures	Processed, organised, and meaningful data
Usefulness on its own	Not directly useful for decision-making	Directly useful — ready to act on
Example	A list of daily temperatures: 31, 33, 29, 35 °C	“The average temperature this week was 32 °C”

Point	Data	Information
Important Point for Exams Remember the one-line definition: data is the INPUT to processing, information is the OUTPUT of processing. Every computer's job, at the simplest level, is to turn data into information.		

2 Characteristics (Advantages) of Computers

Computers are used almost everywhere today because of a specific set of characteristics that make them more capable than humans at certain kinds of work. Each characteristic below is a standard, examinable point.

Characteristic	Explanation	Example
High Speed	A computer can perform calculations on very large amounts of data, in units as small as microseconds, nanoseconds, or even picoseconds — millions of calculations in a few seconds, work that would take a human many months.	Processing a bank's entire day of transactions in seconds
Accuracy	Computers perform every job with 100% accuracy, provided the input given to them is correct.	A calculator never makes an arithmetic mistake
Storage Capability	A computer has far more storage capacity than a human being, and can store any type of data — images, video, text, or audio.	A single hard disk holding a lifetime of family photos
Diligence	Unlike humans, a computer is free from tiredness, monotony, and lack of concentration. It can perform repeated tasks continuously, with the same speed and accuracy every single time.	Running the same payroll calculation for 10,000 employees without a single error
Versatility	A computer is highly flexible — the same machine can solve a complex scientific problem one moment and run a card game the next, simply by loading different software.	Using one laptop for coding, gaming, and video calls
Reliability	Computers are reliable machines; modern electronic components have long working lives, and computers are designed to make maintenance easy.	A server running for months without breaking down
Automation	Once a program is stored in memory, its instructions can control execution without further human interaction — the ability to perform a task automatically.	A washing machine completing a full wash cycle unattended
Reduction in Paper Work & Cost	Using computers for data processing in an organisation reduces paper work and speeds up processes, since electronic files can be retrieved whenever required.	A digital land-record register replacing paper files at a tehsil office

Characteristic	Explanation	Example
Memory Trick for Exams Remember these eight characteristics using: “SAS-DVRA” — Speed, Accuracy, Storage, Diligence, Versatility, Reliability, Automation (plus reduced paperwork/cost). Examiners commonly ask you to “explain any five characteristics with examples” — pick your five strongest and always attach a concrete example to each.		

3 Limitations (Disadvantages) of Computers

Despite their advantages, computers have real limitations that are just as important to understand — and just as commonly examined.

Limitation	Explanation
No I.Q. (No Intelligence)	A computer has no intelligence of its own to decide what to do. Every single instruction must be given to it explicitly; it cannot take any decision on its own beyond what it has been programmed to do.
Dependency	A computer functions purely according to the user's instructions, so it is entirely dependent on humans to tell it what to do and how to do it.
Environment	The operating environment of a computer needs to be dust-free and suitable (stable power, controlled temperature) for it to work reliably.
No Feelings	Computers have no feelings, emotions, or intuition — they cannot judge a situation the way a human can.

Limitation	Explanation
Connecting the Idea — GIGO “No I.Q.” and “Dependency” together explain a very commonly tested idea: GIGO — Garbage In, Garbage Out. Because the computer has no intelligence of its own and depends entirely on the instructions and data it is given, any mistake in the input is carried faithfully through to a wrong (but perfectly, accurately computed) output.	

4 Classification / Types of Computers

Computers are classified in three independent ways: by the technology they use to represent data, by their size and capacity, and by their purpose.

Classification of Computers		
By Technology Used • Digital • Analog • Hybrid	By Size & Capacity • Micro • Mini • Mainframe • Super	By Purpose • General Purpose • Special Purpose

Fig 1.2 — The three classification schemes for computers.

4.1 Classification by Technology Used

According to the technology used to represent and process data, computers are grouped into three types: Digital, Analog, and Hybrid.

4.1.1 Digital Computers

Digital computers are mainly general-purpose computers that represent and store data as discrete quantities or numbers, rather than as a continuously varying signal.

- All processing is done in terms of the numeric (binary digit) representation of data and information.
- Although the user enters data in decimal or character form, the computer internally converts it into binary digits (0s and 1s).
- Because they work with exact, countable values, digital computers give very precise, repeatable results.

Examples

PC (Personal Computer), Laptop, and virtually every modern general-purpose computer.

4.1.2 Analog Computers

Analog computers are special-purpose computers that represent and store data using continuously varying physical quantities such as current, voltage, or frequency — rather than discrete numbers.

- They are programmed for measuring physical quantities like pressure, temperature and speed, and for performing computations directly on those measurements.
- Analog computers are mainly used for scientific and engineering applications.

Examples

Thermometer — a simple analog computer that measures temperature. Speedometer — the position of the needle on the dial represents the car's speed.

Limitations of Analog Computers

1) They do not have the logical facilities that programming affords. 2) They cannot store data in large quantities. 3) The cost of implementing computation on them is very high.

4.1.3 Hybrid Computers

Hybrid computers combine the technology of both analog and digital computers, taking advantage of the strengths of each.

- They store and process analog signals that have been converted into discrete numbers using an analog-to-digital converter.
- They can also convert digital numbers back into analog signals or physical properties using a digital-to-analog converter.
- Hybrid computers are mainly used in artificial intelligence (robotics) and computer-aided manufacturing (e.g., process control).

Examples

Automated Teller Machine (ATM).

Quick Comparison

Feature	Digital	Analog	Hybrid
Represents data as	Discrete numbers (binary)	Continuous physical quantities	Both — converts between the two
Purpose	General-purpose	Special-purpose	Special-purpose
Precision	Very high, exact	Lower, approximate	High, combines both
Typical use	Everyday computing	Scientific / engineering measurement	AI, robotics, process control
Example	PC, Laptop	Thermometer, Speedometer	ATM

4.2 Classification by Size and Capacity

This is the most commonly used classification. Moving from micro to super, computers generally get physically larger, faster, more expensive, and able to serve more users at once — though in modern usage a “micro” computer can still be extremely powerful.

4.2.1 Microcomputers

Microcomputers, also called personal computers (PCs), use a microprocessor as their CPU, along with a memory unit, an input device and an output device.

- They are small in size and do not have large storage capacities.
- They are designed to be used by one person at a time, which is why they are commonly called personal computers.
- Popular uses: word processing, browsing the web, sending/receiving e-mail, spreadsheet calculations, database management, editing photographs, creating graphics, and playing music or games.

Examples

IBM PC, PS/2, Apple II, Macintosh.

4.2.2 Minicomputers

A minicomputer is a multi-user computer, positioned between a microcomputer and a mainframe in power.

- They have higher processing speed and storage capacity than microcomputers.
- Minicomputers can support 4 to 200 users working simultaneously.
- They are used for real-time applications in industries, research centres, etc.

Examples

PDP-11, IBM 8000 series.

4.2.3 Mainframe Computers

Mainframe computers are multi-user, multi-programming, high-performance computers.

- They operate at very high speed, have very large storage capacity, and can handle the combined workload of many users at once.
- They are generally used in centralised databases.
- Mainframes are used in organisations like banks or large companies, where many people need frequent access to the same data.

Examples

CDC 6600, IBM ES/9000 series.

4.2.4 Supercomputers

Supercomputers are the largest and fastest computers in existence.

- A supercomputer has a number of CPUs operating in parallel, which is what makes it so fast.
- They are used for massive data processing and solving highly sophisticated problems — in science and defence, designing and launching missiles, weather forecasting, biomedical research, aircraft design, and automobile design.

Examples

CRAY-3. India has its own series of supercomputers called PARAM, developed by C-DAC, and ANURAG.

Quick Comparison

Feature	Micro	Mini	Mainframe	Super
Users	Single user	4–200 users	Many users (centralised)	Dedicated, specialised jobs
Relative power	Lowest	Higher than micro	Very high	Highest
Typical setting	Home / personal desk	Industry, research centre	Banks, large companies	Science, defence, weather
Example	IBM PC, Macintosh	PDP-11	CDC 6600, IBM ES/9000	CRAY-3, PARAM

4.3 Classification by Purpose

By purpose, computers are grouped into two types depending on whether they are built to do many different jobs, or one specific job.

4.3.1 General Purpose Computers

A general-purpose computer can work on different types of programs given to it, and can therefore be used in countless applications.

- Programs are not permanently stored in the machine; they are input at the time of execution.
- These computers are very versatile.
- Simply by using different software on the same hardware, a general-purpose computer can handle writing and editing (word processing), managing facts in a database, tracking manufacturing inventory, and performing scientific calculations — among many other tasks.

4.3.2 Special Purpose Computers

A special-purpose computer is designed to perform a specific task.

- The instructions (programs) to carry out that task are permanently stored in the machine.
- For its specific task, this type of computer works very efficiently — but such computers are not versatile.
- Useful applications: graphics-intensive video games, traffic-light control systems, navigational systems in aircraft, weather forecasting systems, and satellite launch / tracking systems.

Quick Comparison

Feature	General Purpose	Special Purpose
Programs	Input at the time of execution, easily changed	Permanently stored in the machine
Versatility	Very versatile — handles many tasks	Not versatile — handles one task only
Efficiency at its task	Good, but not optimised for any one job	Extremely efficient at its one job
Example	PC running word processor, spreadsheet, browser, etc.	Traffic-light controller, aircraft autopilot, ATM

Quick Revision Summary

One page to revise everything above the night before a test.

Basis	Type	One-line definition	Example
Technology	Digital	Represents data as discrete numbers (binary); general-purpose.	PC, Laptop
Technology	Analog	Represents data as continuously varying physical quantities.	Thermometer, Speedometer
Technology	Hybrid	Combines analog and digital using A-to-D and D-to-A converters.	ATM
Size	Micro	Personal computer built around a microprocessor; single user.	IBM PC, Macintosh
Size	Mini	Multi-user computer supporting 4–200 users; used for real-time work.	PDP-11
Size	Mainframe	Multi-user, multi-programming, high-performance; centralised database.	CDC 6600, IBM ES/9000
Size	Super	Largest, fastest; many CPUs working in parallel.	CRAY-3, PARAM
Purpose	General Purpose	Runs many different programs, loaded at execution time.	A typical PC
Purpose	Special Purpose	Permanently programmed for exactly one task; not versatile.	ATM, traffic-signal control

Memory hook

Technology asks “how does it store data?” (Digital / Analog / Hybrid). Size asks “how big and how many users?” (Micro < Mini < Mainframe < Super). Purpose asks “one job or many?” (General / Special).

Lecture 1 — Introduction to Computers — the foundation for every later lecture on hardware, software, memory, and networks.